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 Google Scholar profile
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NLP Postdoctoral Research Fellow

Natural Language Processing | Text Simplification | Readability | Explainable AI

Profile

PhD-qualified NLP researcher with over 14 years of experience in computational linguistics, text simplification, readability analysis and applied language technology. My central question is how complex public-facing information can be made easier to understand and use without losing essential meaning. Across the Horizon Europe iDEM project, UCREL and my doctoral research, I develop multilingual corpora, annotation frameworks, simplification pipelines, web-based research software and APIs, and explainable, uncertainty-aware models for accessible and reliable communication.

Research Expertise & Technical Skills

Research focus

Easy-to-Read text; readability and sentence complexity; lexical, syntactic and semantic simplification; accessible communication; multilingual and under-resourced NLP.

Modelling and evaluation

Transformers and LLMs; controlled generation; classification and regression; explainable AI; uncertainty estimation; denoising; error analysis.

Corpus and resource development

Corpus construction, cleaning and alignment; annotation schemes; linguistic feature engineering; Sketch Engine resources; dataset documentation and release.

Programming and data

Python; R; C#; SQL; PyTorch; Hugging Face; scikit-learn; pandas; spaCy; NLTK; Gensim; Oracle databases.

Web and research software

API and web-based NLP tool development; Flask applications; interactive analysis and visualisation tools; .NET web application development.

Infrastructure and reproducibility

Docker; Git/GitHub; Linux and command-line workflows; reproducible experimental pipelines; open-source release and technical documentation.

Selected Research Contributions & Open Outputs

Accessible multilingual simplification iDEM / 2024–present
Lead research on readability, complexity typology and LLM simplification/evaluation. Contributed to multilingual Easy-to-Read resources for democratic participation and meaning-preserving adaptation.

 Democracy Made Easy  E2R corpus paper

Reliable readability modelling 2025–2026

Developed explainability, denoising and uncertainty-estimation analyses that identify why texts are difficult and when a multilingual classifier should abstain from prediction. Released reproducible resources and experimental code.

 why-tough  denoising-difficulty

Arabic simplification and readability PhD / 2021–2023

Developed a corpus-based automatic Modern Standard Arabic text simplification system and released Arabic readability and CEFR-level lexical resources supporting classification, vocabulary profiling and simplification research.

 PhD thesis  Arabic corpus

Software and low-resource language resources UCREL / 2022–2024

Led FreeTxt software prototype development, co-developed an open-source Welsh thesaurus, and contributed to aligned Arabic-English legal corpora.

 FreeTxt  Thesawrws

Research Experience

06/2024 – present	NLP Postdoctoral Research Fellow <i>Horizon Europe – iDEM Project: Innovative and Inclusive Democratic Spaces</i> Lead NLP research on sentence-level readability, complexity classification and text simplification for accessible democratic and institutional information. Develop LLM-based simplification and evaluation pipelines, complexity-typology analysis, interpretability, data denoising and uncertainty-aware assessment; contribute to multilingual original/Easy-to-Read corpora and reusable APIs/resources.	University of Leeds, CTIS
10/2022 – 05/2024	NLP Research Associate <i>FreeTxt, Thesawrws, Welsh Digital Grid and Environmental Change Analysis projects</i> Led software prototype development for FreeTxt, turning bilingual corpus-analysis methods into a web-based toolkit for academic and GLAM-sector users. Served as sole software prototype developer for Thesawrws and co-developed methods for an interactive open-source thesaurus for contemporary Welsh.	Lancaster University, UCREL Group
07/2022 – 2024	Research Assistant <i>Arabic Teaching Council</i> Supported Arabic teaching in Leeds and West Yorkshire through outreach, workshops, professional development events and mentoring coordination.	University of Leeds
03/2022 – 07/2022	Research Assistant – Corpus Builder <i>Arabic-English constitutions corpus project</i> Built, aligned and semi-automatically verified Arabic-English constitutional corpora for publication on Sketch Engine.	University of Leeds

- 05/2020 – 07/2021 **Research Assistant – Corpus Annotator** Lancaster University, Zayed University and NYU Abu Dhabi
Arabic Semantic Annotator based on the UCREL USAS Semantic Tagger
 Researched and annotated a large Arabic semantic lexicon and produced analysis outputs for project presentations.
- 01/2021 – 03/2021 **Research Assistant – Data Analyst** University of Leeds
Arabic learners' perceptions of grammar knowledge
 Conducted qualitative and quantitative analysis of approximately 264 questionnaire responses.
- 10/2020 – 01/2021 **Research Assistant – Tweets Annotator** Lancaster University
Influenza and COVID-19 Arabic Twitter corpus
 Analysed and annotated approximately 2,000 Arabic tweets for public-health discourse categories.
- 12/2010 – 06/2014 **Developer and Corpus Annotator** Alexandria University / RDI
ALTEC diacritised and named-entity annotated corpora
 Developed annotation workflows and interfaces and supported corpus annotation across 104 named-entity categories.

Education

- 2019 – 2023 **PhD in Natural Language Processing** University of Leeds, United Kingdom
Newton–Mosharafa Fund / British Council PhD Research Scholarship
 – Thesis: *An Automatic Modern Standard Arabic Text Simplification System: A Corpus-Based Approach.*
- 2011 – 2016 **MSc in Computational Linguistics** Alexandria University, Egypt
 GPA: 3.8
 – Thesis: *Towards Arabic Named Entity Recognition.*
- 2011 – 2012 **Professional Scholarship in Web Application Development** RAYA Academy for IT / MCIT, Egypt
 Grade: *Excellent*; MCPD and MCTS certifications
- 2006 – 2010 **BA in Phonetics and Linguistics** Alexandria University, Egypt
 Very Good with Honours
 – Graduation project: intelligent online website for teaching Modern Standard Arabic as a second language.

Publications

2026

- Khallaf, N., & Sharoff, S. (2026). *How Much Noise Can BERT Handle? Insights from Multilingual Sentence Difficulty Detection*. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, 6132–6143. ELRA. Proceedings.
- Khallaf, N., & Sharoff, S. (2026). *To Predict or Not to Predict? Towards Reliable Uncertainty Estimation in the Presence of Noise*. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, 152–168. ELRA. Proceedings.
- Riegler, V., Bott, S., Saggion, H., Rascón Alcaina, A., & Khallaf, N. (2026). *A Multilingual Human Annotated Corpus of Original and Easy-to-Read Texts to Support Access to Democratic Participatory Processes*. In *Proceedings of LREC 2026*, 1117–1128. ELRA. Proceedings.

2025

- Hayakawa, A., Khallaf, N., Saggion, H., & Sharoff, S. (2025). *UoL-UPF at TSAR 2025 Shared Task: A Generate-and-Select Approach for Readability-Controlled Text Simplification*. In *Proceedings of TSAR 2025*, 193–210. ACL. ACL Anthology.
- Khallaf, N., Bott, S., Eugeni, C., O'Flaherty, J., Sharoff, S., & Saggion, H. (2025). *Democracy Made Easy: Simplifying Complex Topics to Enable Democratic Participation*. In *Proceedings of AI & EL/PL*, 108–124. EAMT. ACL Anthology.
- Khallaf, N., Ezeani, I., Knight, D., Rayson, P., El-Haj, M., Vidler, J., Davies, J., & Alva-Manchego, F. (2025). *FreeTxt: Analyse and Visualise Multilingual Qualitative Survey Data for Cultural Heritage Sites*. In *Proceedings of RANLP 2025*, 541–545. INCOMA Ltd. ACL Anthology.
- Khallaf, N., Eugeni, C., & Sharoff, S. (2025). *Reading Between the Lines: A Dataset and a Study on Why Some Texts Are Tougher Than Others*. In *Proceedings of WRAICOGS 2025*, 24–34. ICCL. ACL Anthology.

2024–2023

- Knight, D., Khallaf, N., Rayson, P., El-Haj, M., Ezeani, I., & Morris, S. (2024). *FreeTxt: A Corpus-Based Bilingual Free-Text Survey and Questionnaire Data Analysis Toolkit*. *Applied Corpus Linguistics*, 4(3), 100103.
- El-Farahaty, H., Khallaf, N., & Alonayzan, A. (2024). *بناء المدونات اللغوية القانونية الخاصة بجامعة ليدز: منهجيات وتحديات* [Building the University of Leeds Legal Corpora: Methodologies and Challenges]. *Majallat al-Lisaniyat al-Arabiyya*, 18, 96–116.
- Khallaf, N., Arfon, E., El-Haj, M., Morris, J., Knight, D., Rayson, P., Hammouda, T. H., & Jarrar, M. (2023). *Open-Source Thesaurus Development for Under-Resourced Languages: A Welsh Case Study*. In *Proceedings of LDK 2023*, 306–315.
- El-Farahaty, H., Khallaf, N., & Alonayzan, A. (2023). *Building the Leeds Monolingual and Parallel Legal Corpora of Arabic and English Countries' Constitutions: Methods, Challenges and Solutions*. *Corpus Pragmatics*, 7, 103–119.
- Khallaf, N. (2023). *An Automatic Modern Standard Arabic Text Simplification System: A Corpus-Based Approach*. Doctoral thesis, University of Leeds. White Rose eTheses.

Selected Earlier Publications

- Khallaf, N., Sharoff, S., & Soliman, R. (2022). *Towards Arabic Sentence Simplification via Classification and Generative Approaches*. In *Proceedings of WANLP 2022*, 43–52. ACL Anthology.
- El-Haj, M., de Souza, E., Khallaf, N., Rayson, P., & Habash, N. (2022). *AraSAS: The Open-Source Arabic Semantic Tagger*. In *Proceedings of OSACT 2022*.
- Khallaf, N., de Souza, E., El-Haj, M., & Rayson, P. (2022). *Semantic Domains across Topics, Genders, and Languages*. In D. Palfreyman & N. Habash (Eds.), *Bilingual Writers and Corpus Analysis* (pp. 96–120). Routledge.
- Khallaf, N., & Sharoff, S. (2021). *Automatic Difficulty Classification of Arabic Sentences*. In *Proceedings of WANLP 2021*, 105–114. ACL Anthology.
- Khallaf, N., & Alansary, S. (2013). *Towards Arabic Named Entity Recognition System*. In *The Thirteenth Conference on Language Engineering (ESOLEC)*, 217–229. Cairo, Egypt.

Academic Leadership, Research Community & Knowledge Exchange

- **Principal Investigator / Internship Supervisor**, N8 CIR Digital Research Internship Programme: *What Do Large Language Models Know? Text Simplification Analysis* (2025) and selected 2026 project *Interpreting Readability Evaluation with Explainable AI*; supervised research at the interface of accessibility, explainable AI and reproducible NLP.
- **Co-editor**, *Proceedings of the Fourth Workshop on Text Simplification, Accessibility and Readability (TSAR 2025)*, Association for Computational Linguistics.
- **Co-editor**, *Proceedings of the First International Conference on Natural Language Processing and Artificial Intelligence for Cyber Security (NLPACS 2024)*, ACL Anthology.
- **Peer reviewer / programme committee member** for NLP venues; contributor to open science through datasets, GitHub code, annotation frameworks, simplification software and reproducible experiment releases.
- **Teaching/Organising Team**, UCREL Summer School in Corpus Linguistics, Lancaster University (2024): practical sessions on text visualisation and summarisation, Docker for NLP 101, and a mini hackathon/team collaboration challenge.
- **Conference organisation**: Organising Committee, Arabic Linguistics Forum (University of Leeds, 2020); Lead Organiser, LCS Postgraduate Conference (University of Leeds, 2019); Founder/Organiser, Postgraduate Students' Scientific Conference (Alexandria University, 2012 and 2017).

Teaching & Supervision Experience

2024 – 2026	Teaching Contributor <i>MODL5089M Machine Translation and Natural Language Processing (40493)</i> Contributed to teaching Natural Language Processing for translation, supporting student learning in machine translation and NLP concepts and methods.	University of Leeds
2025 – present	Undergraduate Research Supervisor <i>Explainable AI, readability evaluation and text simplification</i> Secured and led two consecutive N8 CIR Summer Internship proposals, supervising undergraduate research on accessibility-oriented NLP.	N8 CIR / University of Leeds
09/2020 – 2022	Teaching Assistant <i>Essential Skills in Practical Arabic</i> Supported language teaching, practical sessions, assessment and student learning in Arabic language provision.	University of Leeds
06/2011 – 09/2018	Demonstrator / Associate Lecturer <i>Phonetics and Linguistics / Language Applications Departments</i> Taught Natural Language Processing, Language Technology, Corpus Linguistics, Applied Linguistics, Specialised Dictionaries and Programming.	Alexandria University
2017 – 2020	Teaching and Student Support Roles <i>Research skills; Arabic as a Foreign Language; academic writing support</i> Delivered research-skills support for postgraduate students and supported Modern Standard Arabic and Egyptian Colloquial Arabic teaching for international learners.	Alexandria University / University of Leeds

Awards & Professional Development

Enrichment Student Community Award Alan Turing Institute, Cohort 2022–2023	Lancaster Summer School in Corpus Linguistics Lancaster University, 2020
SICSS-Edinburgh Summer School University of Edinburgh, 2021	Lisbon Machine Learning School (LxMLS) Instituto Superior Técnico, 2020
Research Software Engineering & Digital Humanities Summer School Alan Turing Institute, 2021	Computational Humanities Summer School & OpenMed Training Heidelberg University / Erasmus+, 2017–2019

Open Software & Data Resources

why-tough: readability explanation dataset Difficulty taxonomy, annotation framework and XAI analysis.  GitHub	denoising-difficulty Multilingual sentence-difficulty denoising experiments and data release.  GitHub
Arabic Readability Corpus & CEFR List Open Arabic sentence-difficulty corpus and CEFR-classified lexical resource.  Corpus  CEFR List	To Predict or Not to Predict Uncertainty-estimation methods and reproducible reliability evaluation.  GitHub
FreeTxt / FreeTxt-Flask Bilingual qualitative survey analysis and visualisation toolkit.  GitHub	Welsh Thesawrws Open-source interactive thesaurus resource for contemporary Welsh.  GitHub